MIL-D-2471D

19 June 1968

SUPERSEDING
MIL-D-2471C
26 May 1964

MILITARY SPECIFICATION

DOUGH DIVIDING MACHINES, AUTOMATIC, POCKET

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers atuomatic dough dividing machines, pocket type.

* 1.2 Classification.- The dough dividing machines shall be of one type of the following sizes and styles as specified (see 6.2):

Size 1 - Two pocket - Capable of dividing not less than 30 pieces of dough per minute.

Size 2 - Four pocket - Capable of dividing not less than 60 pieces of dough per minute.

Style A - Right-hand-side delivery (see 6.6).

Style B - Left-hand-side delivery (see 6.6).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-636 - Motor, Alternating Current, (Fractional Horsepower).

CC-M-641 - Motor, Alternating Current, (Integral Horsepower, 220 HP and Smaller).

FSC 7320

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MILITARY

MIL-B-2316 - Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards and publications required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

BAKING INDUSTRY SANITATION STANDARD (BISSC)

Sanitation Standard No. 8 for Dividers, Rounders and Bun Machines

(Copies of Baking Industry Sanitation Standards Publication may be obtained from the Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, N. Y. 10017.)

UNDERWRITERS' LABORATORIES, INC. (UL)

Electrical Construction Materials List

(Copies may be obtained from Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611, Walt Whitman Road, Melville, L.I., N. Y. 11749, and 1655 Scott Boulevard, Santa Clara, California 95050.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample dough dividing machine covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The dough dividing machine delivered under this specification shall be of the manufacturer's commercially current standard product except for any change necessary to comply with specification requirements.

3.3 Standards.-

3.3.1 UL.- The electrical components shall conform to UL standards, as applicable.

3.3.2 UL certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the electrical components he proposes to furnish under this specification meet UL Standards, as applicable. Acceptable evidence of meeting the UL Standards as applicable, shall be a UL label; a listing in the "Electrical Construction Material List" or a certified test report from a recognized independent testing laboratory, acceptable to the Government indicating that the electrical components of the dough dividing machine conforms to the UL Standards, as applicable.

* 3.3.3 BISSC.- Dough dividing machines shall conform to the applicable requirements of BISSC Standard No. 8.

* 3.3.3.1 BISSC certification.- The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dough dividing machine he proposes to furnish under this specification meets the applicable requirements of BISSC Standard No. 8 as follows:

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(1) Display of the BISSC seal on the finished dough dividing machine shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment.

-or-

(2) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service of which the dough dividers are being procured, indicating that the dividers have been tested and conform to BISSC Standard No. 8 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer in fabricating his standard commercial dough dividing machine.

* 3.5 Design.- The dough dividing machines shall be of the pocket type. The machines shall consist of a dough hopper, a dough ram which will force the dough into the pockets, a scaling device for weighing dough pieces, a duster which will dust the dough pieces and two belts which will allow transfer of the dough during the dividing operation. The size 1 machine shall be capable of producing not less than 30 pieces of dough per minute, and the size 2 machine shall be capable of producing not less than 60 pieces of dough per minute when tested as specified in 4.5.1.1.

* 3.6 Construction.- The dough dividing machines shall be constructed as specified herein. The machine shall have a cast iron or fabricated steel frame; and all moving parts shall be fully enclosed. Supports for all shafts shall have replaceable bearings. Vital moving parts shall be protected by shear link, shear pins, or friction safety device.

* 3.6.1 Dough hopper.- The dough hopper shall be constructed of cast iron, welded steel, or aluminum. The distance from the floor to the top of the hopper, when in loading position, shall not exceed 72 inches. The hopper volume shall be 1-1/2 cubic feet for the two-pocket machine, and 2 cubic feet for the four-pocket machine with a tolerance of plus or minus 1/2 cubic foot for each size machine.

3.6.2 Dough ram.- Each dough ram shall have an automatic compensator or an adjustable spring controlled loader to exert the pressure necessary for securing uniform scaling throug.. the full range of the pocket capacity.

3.6.3 Scaling range.- The scaling range of the dividers shall be from 8 ounces to at least 36 ounces of dough for each pocket. Each pocket shall have an individual adjustment for volumetric capacity which will permit all pockets to scale uniformly. The scaling capacity of all pockets shall be regulated by means of a hand adjustment.

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3.6.3.1 Accuracy.- The accuracy of the scaling mechanism shall be plus or minus 5 percent and shall be tested in accordance with the settings specified in 4.5.1.2.

3.6.4 Pockets.- The pockets shall be either circular or rectangular in cross section.

3.6.5 Dividing heads.- Each dividing head shall be of the vertically reciprocating or rotary oscillating type. The pistons and the dividing head shall each be marked to identify each piston with its proper cylinder.

3.6.6 Conveyor belts.-

* 3.6.6.1 Primary belt.- The primary belt below the dividing head shall be arranged to allow transporting the pieces of dough to the right for the style A machine and to the left for the style B machine.

3.6.6.2 Secondary belt.- A secondary belt shall be provided with brackets to permit adjustment for various heights at the delivery point to the rounder. The delivery point shall be not less than 6 feet nor more than 9 feet from the vertical centerline of the divider. The secondary canvas belt shall be provided with cleats at intervals not exceeding 6 inches. The cleats shall be affixed to the belt with nonmetallic fasteners. A polychloroprene rough top belt may be used as an alternate.

3.6.7 Dusters.- Adjustable dusters, driven by the divider mechanism, shall apply flour to the conveyor belts. Removable pans shall be provided under the full length of the two conveyor belts to receive the waste flour, except that a pan shall not be required under the secondary belt when the belt discharges waste flour into the rounder.

3.6.8 Drive unit.- The drive unit shall consist of a power unit which shall provide the operating power through a variable speed transmission to the countershaft or reduction gears.

3.6.9 Lubrication.- Lubrication for all parts of the machine that come in contact with dough shall be provided by an automatic system or sight feed adjustable oilers designed for the use of mineral oil. Hydraulic type grease fittings shall be provided for all grease lubricated bearings. If the unit requires the use of several types of oil, a noncorrosive metal tag, identifying the lubricant to be used shall be permanently attached to each fitting. Seals shall be provided where necessary to prevent leakage of lubricant into the product zone.

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3.6.9.1 Lubrication chart.- The supplier shall affix to each machine a corrosion-resistant metal lubricating chart indicating:

(a) All points of the equipment requiring lubrication.

(b) Grade of lubricant required.

(c) Time interval when lubrication should be accomplished.

* 3.6.10 Electrical requirements.-

* 3.6.10.1 Electric motors.- Electric motors shall conform to CC-M-636 or CC-M-641, as applicable. Motors shall be of the continuous-duty, induction type, totally inclosed, without commutation devices, having not more than one starting contact. The motors shall have ample power to operate the machine and components continuously at maximum capacity without exceeding the allowable temperature rise. The dough dividing machine shall be completely wired and equipped with a mounted pushbutton station and magnetic switch with thermal overload protection. The pushbutton switch shall be located at the front of the machine.

* 3.6.10.2 Electric rating.- Unless otherwise specified (see 6.2), all electrical equipment shall be designed for operation on a nominal 208-volt, alternating current (AC), 60-cycle, three-phase power system.

* 3.7 Electromagnetic compatibility.- Equipment procured for Army shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.5.3). When specified (see 6.2), equipment procured for Navy and Air Force shall be in accordance with class IIB of MIL-STD-461 (see 4.5.3).

3.8 Finish.- All exterior surfaces of the machine not coming in contact with the dough, except corrosion-resisting steel shall be given a white enamel finish in accordance with the manufacturer's standard commercial practice.

3.9 Operation.- The following operational determinations shall be made when the dough divider is tested as specified in 4.5.2.

(a) Operation of drive unit, correct rotation.

(b) Smoothness of operation of variable speed drive while being operated throughout the entire range of adjustment.

(c) Smoothness of operation of the pocket adjustment control throughout the entire range of adjustment without binding.

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(d) Proper operation of all sight feed adjustable oilers or automatic oiling devices.

(e) Proper operation of the conveyor belt in the direction specified (see 3.6.6.1).

3.10 Marking for identification.- Identification marking shall be on a corrosion-resisting steel plate in accordance with MIL-STD-130. Any additional pertinent rating characteristics or information necessary for safe handling operations and maintenance of the item, if any, shall be entered.

* 3.11 Manuals.- The supplier shall furnish an equipment instruction manual with each dough dividing machine. Manual contents and additional quantities required shall be as specified in the contract or order (see 6.2).

* 3.12 Repair and maintenance.- Provisions shall be made to allow for ready adjustment, service and replacement of components without disturbing or disassembly of any major components. In addition there shall be free access to inspect, service, and adjust any component or equipment as provided for by this specification or as indicated by the component or equipment manufacturer, without requiring disassembly of any other component or equipment on the dough dividing machine.

3.13 Workmanship.-

3.13.1 Fabrication.- Metal used shall be free from fractures, splits, punctures, dents, deterioration and shall not be bowed, or malformed. There shall be no sharp burr, sliver or splinter.

3.13.2 Welding and brazing.- All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures, undercuts, or incomplete fusion. All scale or flux deposits shall be removed from finished welds. Welds shall be smooth and uniform.

3.13.3 Solder.- All surfaces to be soldered (when applicable) shall be thoroughly cleaned prior to soldering. Solder joints shall be smooth with no pin holes.

3.13.4 Threaded fasteners.- Threaded fasteners shall not be loose or missing. Threads shall not be missing, stripped or broken.

3.13.5 Electrical assembly.- Wiring shall (1) be properly joined, (2) not be cut or abraded, (3) not be loose at terminal end, (4) be properly inclosed in conduit when required, (5) be provided with slack, and (6) not be stripped of required insulation.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I, dimensions specified and tested as specified in 4.5.1.1, 4.5.1.2, 4.5.2 and 4.5.3.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Process examination.- Examination shall be made during the manufacturing process to insure that the components are cleaned prior to operation (see 3.13.2 and 3.13.3) to establish that no deviation is made from the indicated requirements. Whenever nonconformance is noted, correction or replacement shall be made to affected items and process.

4.3.3 End item inspection.- A lot shall consist of all dough dividing machines of the same size and style offered for inspection at one time. The sample unit shall be one completely fabricated dough dividing machine.

* 4.3.3.1 Visual examination.- The end item shall be examined for defects set forth in table I. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

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TABLE I.- Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not as specified		X
Design	Any characteristic not in accordance with the specified requirements	X	
Construction and workmanship general (applicable to all components and assemblies)	Part missing, or not specified size	X	
	Fractured, split, punctured dented, deteriorated, bowed malformed	X	
	Part misplaced, or not in proper alignment	X	
	Sharp burr, sliver, or splinters	X	
	Any component loose		X
	Functioning component that requires abnormal force to operate		X
	Driving mechanism not inclosed	X	
	Not connected, or joined as specified	X	
	Vital moving parts not protected by shear links, or shear pins, or friction saving device	X	
	Any adjustable assembly that cannot be adjusted	X	
Welding and brazing (when applicable)	Missing, incomplete, burn hole, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth, and uniform, scale or flux deposits not removed		X
Soldering (when applicable)	Missing, not adherent, or incomplete	X	
	Not clean (flux residue not removed), not smooth (surface not finished neatly), or pin holes in solder		X
Castings (when applicable)	Blowhole, porosity shrinkage, or cracks	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws studs and other types of threaded fasteners	Not type specified, broken stripped, fractured, missing, or loose		X
Rivets and pins	Broken not peened, or insufficiently peened		X
Conveyor belts	Missing, not type specified	X	
Electrical apparatus and wiring	Wiring cut, abraded insulation stripped, not properly joined, loose at terminal, not inclosed in conduit where required	X	
	Adequate slack not provided for wiring to relieve strain		X
Scaling range	Not specified capacity	X	
Pocket	Not individual adjustment	X	
Controls	Not specified type	X	
Lubrication	Lubrication system, or fitting not provided, or not type specified	X	
	Insufficiently lubricated		X
	Lubrication chart missing		X
Assembly	Component not properly assembled		X
	Unit perceptibly out of square or alignment	X	
Identification marking	Missing, incomplete, not legible, not specified type		X

4.3.3.2 Dimensional examination.- Inspection shall be made of the sample unit to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be level S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

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4.3.4 Testing of the end item.- When a preproduction sample is not required the first production unit shall be tested as specified in 4.5.1 and 4.5.3 (see 6.2). Each dough dividing machine shall be tested as specified in 4.5.2. Any nonconformance shall be cause for rejection.

4.3.5 Examination of preparation for delivery.- Examination shall be made for preservation, packaging, packing and marking in accordance with MIL-B-2316.

4.4 Code and standard compliance.- The supplier shall have available for the Government representative proof of compliance with the requirements of 3.3.1.

4.5 Tests.-

4.5.1 Functional test.-

4.5.1.1 Rate.- The dough dividing machine shall be connected to power and operated for 15 minutes to determine the rate of pieces of dough for compliance with 3.5.

4.5.1.2 Accuracy.- The dough dividing machine shall be connected to power and operated at dough pocket settings of 10, 20 and 30 ounces to determine compliance with 3.6.3.1.

4.5.2 Operational.- Each dough dividing machine shall be connected to power and operated without load for a minimum of 15 minutes, to determine compliance with 3.9.

4.5.3 Electromagnetic compatibility test.- When electromagnetic compatibility is required, the preproduction sample or initial unit of production shall be tested by the supplier in accordance with test method CE03, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer for approval the Interference Control Plan, the EMI/EMC Test Plan and the Electromagnetic Compatibility Test Report required in MIL-STD-461. Upon approval of the test report by the contracting officer and provided all other requirements of the specification are met, the preproduction sample or initial unit of production as applicable, shall be used as a model for all production units.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C.- Preservation, packaging, packing, and marking for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

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6. NOTES

6.1 Intended use.- The dough dividing machines covered by this spec f ation are intended for use in garrison bakeries.

6.2 Ordering data.- Procurement documents should specify the following

(a) Title, number, and date of this specification.

(b) Size and style machine required (see 1.2).

(c) First article.- When preproduction sample is not required (see 3.1).

(d) Electrical characteristic if other than specified (see 3.6.10.2).

(e) When electromagnetic compatibility is required for Navy and Air Force use (see 3.7).

(f) Manual contents and quantity required (see 3.11).

(g) Selection of the applicable levels of preservation and packaging and packing (see 5.1).

(h) Whether supervision of installation is to be provided (see 6.4).

6.4 Supervision.- Supervision of installation should be required of the supplier on dough dividing machine(s) shipped to any point in Continental U.S., except those dough dividing machines designated for storage or warehouse stock (see 6.2). When supervision of installation is required, the supplier should furnish the services of a competent superintendent who shall be familiar with the construction, erection, operation, maintenance and repair of the equipment and the required utility connections. Material required for complete installations other than that specified herein and all installation labor should be furnished by the Government. All utilities should be extended and connected to the equipment by the Government.

* 6.5 Electromagnetic compatibility.- The contracting officer should submit a copy of the Interference Control Plan, the EMI/EMC Test Plan, and the Test Report required (see 4.5.3) to the Commanding General, U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey for approval.

6.6 The term right-hand-side delivery indicates, with the operator viewing the dough divider from the front and the conveyor is extending out to the right. The term left-hand-side delivery indicates, with the operator viewing the dough divider from the front and the conveyor is extending out to the left.

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* 6.7 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Review activities:

Army - EL, MD
Navy - MC, YD
Air Force - 84, 85

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Dough Dividing Machines, Automatic, Pocket MIL-D-2471D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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